Imperial-Cambridge Math Contest 2022 — P1/6

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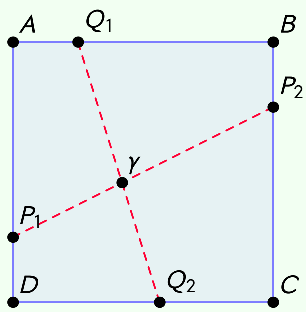
Two straight lines divide a square of side length 1 into four regions. Show that at least one of the regions has a perimeter greater than or equal to 2.

Solution

Let Q_1Q_2 and P_1P_2 divide the square $ABCD$ into four regions, and let γ be the intersection of P_1P_2 and Q_1Q_2.



If P_1P_2 and Q_1Q_2 both intersect opposite sides of the square then they both will have length ≥ 1. Now the total perimeter P of the four regions will be

$$P = 4 + 2(Q_1\gamma + Q_2\gamma + P_1\gamma + P_2\gamma) \geq 4 + 2(1 + 1) = 8.$$

Now the region with the largest perimeter must have perimeter ≥ 2, because otherwise the total perimeter would be $P < 2 + 2 + 2 + 2 = 8$, a contradiction.

Now if exactly one of these lines intersect opposite sides of the square, say WLOG P_1P_2. Now WLOG let Q_1Q_2 be such that Q_1 lies on the opposite side of whatever side P_2 lies on. Then one of the four regions is bounded by $P_2\gamma$, $Q_1\gamma$ and one of the sides of the square. Notice that $P_2\gamma + Q_1\gamma \geq P_2Q_1 \geq 1$. The perimeter of this region is $\geq 1 + 1 = 2$.

Now finally if none of the lines intersect opposite sides of the square then one of the regions includes two adjacent sides of the square, which means the perimeter of that region is at least $1 + 1 = 2$. ■